

initiated the action. This information can be very useful when tracking down runaway processes and threads that hide themselves deep within the system and the Registry.

3.3 Registry Cleaners

One of the inherent problems with the Windows Registry is the number of obsolete entries it will accumulate over the life of the system. Applications that are uninstalled according to Microsoft's guidelines are required to remove themselves from the Registry as well. However, not all applications follow these guidelines, and more often than not, traces of the application will be left over in the Registry. This might not pose much of a problem in the beginning, but over time—especially if you are prone to installing and uninstalling a lot of applications—the pile of obsolete entries will begin to grow, and there will be a performance impact on the Registry, which in turn will impact the overall performance of the system. Also, if the software's uninstall program is sloppy, some Registry references may create problems for other installed applications or fresh software installations.

Registry cleaners attempt to solve this problem by scanning the Registry to fix and correct errors. These could include things like missing references to shared DLLs, unused file extensions, ActiveX and Class issues, obsolete software, shortcut paths, and more. There are no clear guidelines on how to effectively clean the Registry; as a result, developers of Registry cleaning software attempt their own methods. Some take an aggressive approach and attempt to clean everything that looks remotely obsolete or bad, which in some cases can do more harm than good. Others go for the middle path and exclude cleaning out entries for which there is some doubt.

A cleaned registry will improve system performance, since obsolete and erroneous entries will no longer slow down access to the Registry—thus improving application response time.